Humanoid Robot Modeling and Dynamics

Quaternion floating-base formulation

From kinematic trees and Jacobians to the equations of motion and contact-constrained motion

Technical study notes | Floating-base rigid-body model | Rewritten from the Euler-angle version: the base orientation is a unit quaternion and the base velocity is the physical angular velocity.

1. Kinematic tree and floating base

A humanoid consists of rigid links (nodes) connected by joints (edges). The pelvis is selected as the root of a kinematic tree. Each link stores its mass, center of mass, inertia tensor, parent link, joint axis and fixed transforms. A preorder traversal (parents before children) computes parent-to-child world transforms.

Configuration and generalized velocity

The configuration and the generalized velocity are two different vectors of two different sizes:

$$\mathbf{q} = \begin{bmatrix} \mathbf{p}_b \\ \boldsymbol{\xi}_b \\ \mathbf{q}_j \end{bmatrix} \in \mathbb{R}^{n+7}, \qquad \boldsymbol{\nu} = \begin{bmatrix} \dot{\mathbf{p}}_b \\ \boldsymbol{\omega}_b \\ \dot{\mathbf{q}}_j \end{bmatrix} \in \mathbb{R}^{n+6}. \tag{1}$$

Here $\mathbf{p}_b \in \mathbb{R}^3$ is the base (pelvis) position, $\boldsymbol{\xi}_b = (w, x, y, z) = (w, \boldsymbol{\epsilon})$ is a unit quaternion ($\|\boldsymbol{\xi}_b\| = 1$), $\mathbf{q}_j$ contains the n actuated joint coordinates, and $\boldsymbol{\omega}_b$ is the angular velocity of the base. In these notes $\dot{\mathbf{p}}_b$ and $\boldsymbol{\omega}_b$ are both expressed in the *world* frame. The base has six unactuated degrees of freedom: 6 velocity components, stored with 7 configuration numbers.

Because the quaternion has 4 components but only 3 rotational degrees of freedom, $\boldsymbol{\nu} \neq \dot{\mathbf{q}}$. They are related by a configuration-dependent map:

$$\dot{\mathbf{q}} = N(\mathbf{q})\,\boldsymbol{\nu}, \qquad N(\mathbf{q}) = \begin{bmatrix} I_3 & 0 & 0 \\ 0 & \frac{1}{2}H(\boldsymbol{\xi}_b) & 0 \\ 0 & 0 & I_n \end{bmatrix} \in \mathbb{R}^{(n+7)\times(n+6)}, \qquad H(\boldsymbol{\xi}) = \begin{bmatrix} -\boldsymbol{\epsilon}^{\mathsf{T}} \\ wI_3 - [\boldsymbol{\epsilon}]_\times \end{bmatrix} \in \mathbb{R}^{4\times 3}. \tag{2}$$

Equivalently $\dot{\boldsymbol{\xi}}_b = \frac{1}{2}\,(0, \boldsymbol{\omega}_b) \otimes \boldsymbol{\xi}_b$, which multiplies on the *left* because $\boldsymbol{\omega}_b$ is in the world frame. With a body-frame angular velocity it would instead be $\frac{1}{2}\,\boldsymbol{\xi}_b \otimes (0, \boldsymbol{\omega}_b)$.

Rotations and transforms

The rotation matrix of a unit quaternion is

$$R(\boldsymbol{\xi}) = \begin{bmatrix} 1-2(y^2+z^2) & 2(xy-wz) & 2(xz+wy) \\ 2(xy+wz) & 1-2(x^2+z^2) & 2(yz-wx) \\ 2(xz-wy) & 2(yz+wx) & 1-2(x^2+y^2) \end{bmatrix}, \tag{3}$$

which is the Rodrigues formula $R = I + \sin\theta\,[\mathbf{a}]_\times + (1-\cos\theta)[\mathbf{a}]_\times^2$ written with $w = \cos\frac{\theta}{2}$, $\boldsymbol{\epsilon} = \sin\frac{\theta}{2}\,\mathbf{a}$. Note that $\boldsymbol{\xi}$ and $-\boldsymbol{\xi}$ give the same rotation. A revolute joint with local axis $\mathbf{a}$ and angle q_i uses the same Rodrigues formula directly. World transforms are composed down the tree:

$${}^{W}T_i(\mathbf{q}) = {}^{W}T_{\mathrm{parent}(i)}(\mathbf{q})\;{}^{\mathrm{parent}(i)}T_{\mathrm{origin},i}\;\mathrm{Rot}(\mathbf{a}_i, q_i), \qquad {}^{W}T_b = \begin{bmatrix} R(\boldsymbol{\xi}_b) & \mathbf{p}_b \\ 0 & 1 \end{bmatrix}. \tag{4}$$

For each link i we write $(R_i, \mathbf{o}_i)$ for the world rotation and frame origin, $\mathbf{c}_i$ for the center of mass in the link frame, and $\mathbf{p}_i = \mathbf{o}_i + R_i\mathbf{c}_i$ for the world position of its center of mass.

2. Link Jacobians

The translational and angular Jacobians map the generalized *velocity* $\boldsymbol{\nu}$ (not $\dot{\mathbf{q}}$) to the center-of-mass linear velocity and the link angular velocity:

$$\mathbf{v}_i = J_{v_i}(\mathbf{q})\,\boldsymbol{\nu}, \qquad \boldsymbol{\omega}_i = J_{\omega_i}(\mathbf{q})\,\boldsymbol{\nu}, \qquad J_{v_i}, J_{\omega_i} \in \mathbb{R}^{3\times(n+6)}. \tag{5}$$

Every column follows from the rigid-body velocity rule $\dot{\mathbf{x}} = \dot{\mathbf{o}} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{o})$.

Floating-base columns. With $\boldsymbol{\nu}$ as defined above they are the same for every orientation, and contain no quaternion terms:

$$J_{v_i}^{(\text{base})} = \begin{bmatrix} I_3 & -[\mathbf{p}_i - \mathbf{p}_b]_\times \end{bmatrix}, \qquad J_{\omega_i}^{(\text{base})} = \begin{bmatrix} 0_3 & I_3 \end{bmatrix}. \tag{6}$$

Joint columns. For a revolute ancestor joint j with world axis $\mathbf{a}_j$ and axis point $\mathbf{p}_j$:

$$J_{v_i}^{(j)} = \mathbf{a}_j \times (\mathbf{p}_i - \mathbf{p}_j), \qquad J_{\omega_i}^{(j)} = \mathbf{a}_j. \tag{7}$$

For a prismatic ancestor, $J_{v_i}^{(j)} = \mathbf{a}_j$ and $J_{\omega_i}^{(j)} = 0$. Columns of joints that are not ancestors of link i are zero.

Jacobian time derivative. This is needed in Sections 6 and 7. Since the axis is fixed in the moving link, $\dot{\mathbf{a}}_j = \boldsymbol{\omega}_j \times \mathbf{a}_j$, where $\boldsymbol{\omega}_j$ is the angular velocity of the joint's child link. The columns differentiate as

$$\tfrac{d}{dt} J_{v_i}^{(j)} = \dot{\mathbf{a}}_j \times (\mathbf{p}_i - \mathbf{p}_j) + \mathbf{a}_j \times (\mathbf{v}_i - \dot{\mathbf{p}}_j), \qquad \tfrac{d}{dt} J_{\omega_i}^{(j)} = \dot{\mathbf{a}}_j, \qquad \tfrac{d}{dt} J_{v_i}^{(\text{base})} = \begin{bmatrix} 0 & -[\mathbf{v}_i - \dot{\mathbf{p}}_b]_\times \end{bmatrix}. \tag{8}$$

In practice one only needs the product $\dot{J}\boldsymbol{\nu}$, which is the acceleration of the point when $\dot{\boldsymbol{\nu}} = 0$ (the "bias acceleration").

3. Target dynamics equation

$$M(\mathbf{q})\,\dot{\boldsymbol{\nu}} + C(\mathbf{q},\boldsymbol{\nu})\,\boldsymbol{\nu} + G(\mathbf{q}) = S^{\mathsf{T}}\boldsymbol{\tau} + J_c(\mathbf{q})^{\mathsf{T}}\mathbf{f}_c, \qquad \mathbf{h}(\mathbf{q},\boldsymbol{\nu}) := C(\mathbf{q},\boldsymbol{\nu})\boldsymbol{\nu} + G(\mathbf{q}) \tag{9}$$

Quantity	Meaning
$M(\mathbf{q}) \in \mathbb{R}^{(n+6)\times(n+6)}$	Symmetric positive-definite generalized mass matrix
$C(\mathbf{q},\boldsymbol{\nu})\boldsymbol{\nu}$	Velocity-dependent Coriolis, centrifugal and gyroscopic vector
$G(\mathbf{q})$	Gravity contribution, written on the left-hand side
$S^{\mathsf{T}}\boldsymbol{\tau}$	Generalized joint actuation; $S = [\,0_{n\times 6} \;\; I_n\,]$
$J_c^{\mathsf{T}}\mathbf{f}_c$	Generalized forces due to contacts
$\dot{\boldsymbol{\nu}}$	Generalized acceleration ($\neq \ddot{\mathbf{q}}$, which would have $n+7$ entries)

All matrices act on the $(n+6)$-dimensional velocity space. The quaternion appears only inside M, C, G and J_c through $R(\boldsymbol{\xi}_b)$, and in the integration step (Section 9).

4. Construct kinetic energy from Jacobians

The kinetic energy of each rigid link includes translation of its center of mass and rotation about its center of mass. $I_i^W = R_i\, I_i^{\text{body}} R_i^{\mathsf{T}}$ is the world-frame inertia tensor about the center of mass.

$$T_i = \tfrac{1}{2} m_i \mathbf{v}_i^{\mathsf{T}}\mathbf{v}_i + \tfrac{1}{2}\boldsymbol{\omega}_i^{\mathsf{T}} I_i^W \boldsymbol{\omega}_i = \tfrac{1}{2}\,\boldsymbol{\nu}^{\mathsf{T}}\left[m_i J_{v_i}^{\mathsf{T}} J_{v_i} + J_{\omega_i}^{\mathsf{T}} I_i^W J_{\omega_i}\right]\boldsymbol{\nu}. \tag{10}$$

Summing over all N links:

$$M(\mathbf{q}) = \sum_{i=1}^{N}\left[m_i J_{v_i}^{\mathsf{T}} J_{v_i} + J_{\omega_i}^{\mathsf{T}} I_i^W J_{\omega_i}\right], \qquad T(\mathbf{q},\boldsymbol{\nu}) = \tfrac{1}{2}\,\boldsymbol{\nu}^{\mathsf{T}} M(\mathbf{q})\,\boldsymbol{\nu}. \tag{11}$$

This is unchanged from the Euler-angle version, except that the Jacobians now map $\boldsymbol{\nu}$, which makes M simpler: no Euler-rate matrix $E(\boldsymbol{\theta}_b)$ appears in it. M is independent of $\mathbf{p}_b$. With world-frame base velocity it depends on $\boldsymbol{\xi}_b$ only through the world inertias I_i^W.

5. Gravitational potential energy

Choose the world z-axis upward and $g > 0$ as the magnitude of gravitational acceleration. Let $z_i(\mathbf{q})$ be the height of the center of mass of link i, so that $V(\mathbf{q}) = \sum_i m_i g\, z_i(\mathbf{q})$.

With a quaternion, $\partial V/\partial \mathbf{q}$ has $n+7$ entries and is not the generalized force. The generalized force is defined by *virtual power*: the rate of change of V along a velocity $\boldsymbol{\nu}$.

$$\dot{V} = \sum_i m_i g\, \mathbf{e}_z^{\mathsf{T}} \mathbf{v}_i = \Big(\sum_i m_i g\, J_{v_i}^{\mathsf{T}} \mathbf{e}_z \Big)^{\mathsf{T}} \boldsymbol{\nu} \quad \Longrightarrow \quad G(\mathbf{q}) = \sum_i m_i g\, J_{v_i}^{\mathsf{T}} \mathbf{e}_z, \qquad \mathbf{e}_z = [0,0,1]^{\mathsf{T}}. \tag{12}$$

This is the same formula as before; $G = N(\mathbf{q})^{\mathsf{T}}\, \partial V/\partial \mathbf{q}$. The generalized gravitational force is $-G(\mathbf{q})$, and $G(\mathbf{q})$ is written on the left-hand side. For the base translation rows, $G_{0:3} = (\sum_i m_i)\, g\, \mathbf{e}_z$, the total weight.

6. Derivation of the equations of motion

Why the plain Euler–Lagrange equation needs care

The classical form $\frac{d}{dt}(\partial L/\partial \dot{\mathbf{q}}) - \partial L/\partial \mathbf{q} = Q$ assumes the velocity is the time derivative of independent coordinates. Here $\boldsymbol{\omega}_b$ is not the derivative of any coordinate (it is a *quasi-velocity*), and the 4 quaternion components are not independent. The Christoffel formula for C from the Euler-angle notes therefore cannot be applied to $\boldsymbol{\nu}$ directly. Two equivalent routes are available: Lagrange's equations in quasi-velocities (Boltzmann–Hamel), or d'Alembert's principle of virtual power (Kane's method). The second is shorter and is used here; both give the same M, $C\boldsymbol{\nu}$ and G.

Step 1 — Newton–Euler equations of each link

For link i with external force $\mathbf{F}_i$ at the center of mass and external moment $\mathbf{N}_i$ (world frame, about the center of mass):

$$m_i \dot{\mathbf{v}}_i = \mathbf{F}_i, \qquad I_i^W \dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i^W \boldsymbol{\omega}_i = \mathbf{N}_i. \tag{13}$$

The gyroscopic term comes from $\frac{d}{dt}(I_i^W \boldsymbol{\omega}_i)$ with $\dot{I}_i^W = [\boldsymbol{\omega}_i]_\times I_i^W - I_i^W [\boldsymbol{\omega}_i]_\times$.

Step 2 — Virtual power

For every admissible virtual velocity $\boldsymbol{\nu}^*$, with $\mathbf{v}_i^* = J_{v_i}\boldsymbol{\nu}^*$ and $\boldsymbol{\omega}_i^* = J_{\omega_i}\boldsymbol{\nu}^*$, the inertial and applied forces balance. Internal joint reaction forces do no work and drop out:

$$\sum_i \left[J_{v_i}^{\mathsf{T}} \left(m_i \dot{\mathbf{v}}_i \right) + J_{\omega_i}^{\mathsf{T}} \left(I_i^W \dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i^W \boldsymbol{\omega}_i \right) \right] = Q, \tag{14}$$

where Q collects gravity, actuation and contact (Step 4).

Step 3 — Substitute the accelerations

Differentiating the Jacobian relations gives $\dot{\mathbf{v}}_i = J_{v_i}\dot{\boldsymbol{\nu}} + \dot{J}_{v_i}\boldsymbol{\nu}$ and $\dot{\boldsymbol{\omega}}_i = J_{\omega_i}\dot{\boldsymbol{\nu}} + \dot{J}_{\omega_i}\boldsymbol{\nu}$. Collecting the $\dot{\boldsymbol{\nu}}$ terms reproduces M from Section 4, and the remaining terms give the Coriolis matrix:

$$C(\mathbf{q}, \boldsymbol{\nu}) = \sum_i \left[m_i J_{v_i}^{\mathsf{T}} \dot{J}_{v_i} + J_{\omega_i}^{\mathsf{T}} \left(I_i^W \dot{J}_{\omega_i} + [\boldsymbol{\omega}_i]_\times\, I_i^W J_{\omega_i} \right) \right]. \tag{15}$$

With this choice $\dot{M} - 2C$ is skew-symmetric, the same passivity property as the Christoffel form. It is useful for checking and for control design.

Step 4 — Generalized forces

Gravity on each link, $\mathbf{F}_i = -m_i g\, \mathbf{e}_z$, contributes $-G(\mathbf{q})$ (Section 5). A torque τ_k about joint axis k, acting equally and oppositely on parent and child, has virtual power $\tau_k \dot{q}_k^*$, so actuation contributes $S^\mathsf{T}\boldsymbol{\tau}$. A contact force $\mathbf{f}_c$ at a point with Jacobian J_c has virtual power $\mathbf{f}_c^\mathsf{T} J_c \boldsymbol{\nu}^*$, contributing $J_c^\mathsf{T}\mathbf{f}_c$. Therefore:

$$M(\mathbf{q})\dot{\boldsymbol{\nu}} + C(\mathbf{q},\boldsymbol{\nu})\boldsymbol{\nu} + G(\mathbf{q}) = S^\mathsf{T}\boldsymbol{\tau} + J_c^\mathsf{T}\mathbf{f}_c. \tag{16}$$

The first 6 rows (the base rows, with no actuator) are the Newton–Euler equations of the whole robot: the total momentum changes only through gravity and contact forces.

7. Foot contact modes and constraints

Swing foot

A non-contacting foot has zero contact force and no sticking constraint. Other active contacts can still provide support.

Sticking point contact

For a point contact at $\mathbf{p}_c(\mathbf{q})$ fixed at $\mathbf{p}_c^*$, impose $\mathbf{p}_c(\mathbf{q}) - \mathbf{p}_c^* = 0$. Differentiation along the motion gives velocity- and acceleration-level constraints in terms of $\boldsymbol{\nu}$:

$$J_c(\mathbf{q})\,\boldsymbol{\nu} = 0, \qquad J_c(\mathbf{q})\,\dot{\boldsymbol{\nu}} + \dot{J}_c(\mathbf{q},\boldsymbol{\nu})\,\boldsymbol{\nu} = 0, \tag{17}$$

with J_c the $3 \times (n+6)$ point Jacobian from Section 2 (evaluated at the contact point instead of the center of mass). A sticking point contact imposes three translational constraints; it does not directly fix foot orientation. For the simple biped, four point contacts at the sole corners of each foot are used. Two points on the foot centerline (heel and toe only) cannot resist rolling about that line.

Fixed flat-foot contact

A rigid foot fully fixed to the ground has six local pose constraints: three translational and three rotational. Its contact Jacobian stacks the linear and angular foot Jacobians:

$$J_c = \begin{bmatrix} J_{v_f} \\ J_{\omega_f} \end{bmatrix} \in \mathbb{R}^{6\times(n+6)}. \tag{18}$$

The rotational rows constrain the angular velocity $\boldsymbol{\omega}_f$ directly, so no quaternion or Euler-angle derivative is involved. A flat sole by itself does not imply no slip or fixed yaw; those are additional contact assumptions.

Tilted foot and inclined terrain

A foot tilted relative to horizontal ground may contact at a toe, heel or edge, depending on geometry. A foot aligned with a sloped plane can be flat relative to that plane even though tilted in world coordinates. Active constraints depend on actual geometry and friction.

Unilateral contact and friction

For gap $\phi(\mathbf{q})$ from the surface and normal force λ_n, the complementarity and Coulomb sticking conditions are:

$$\phi(\mathbf{q}) \geq 0, \quad \lambda_n \geq 0, \quad \phi(\mathbf{q})\,\lambda_n = 0; \qquad \|\mathbf{f}_t\| \leq \mu\lambda_n. \tag{19}$$

These conditions prevent penetration and tensile (pulling) normal contact; insufficient friction may cause slip. A simpler alternative for a first simulator is a penalty (spring–damper) model, $\lambda_n = \max(0,\, -k\phi - d\dot{\phi})$, which turns contact into an ordinary applied force.

8. Constrained dynamics linear system

For a fixed active set of independent sticking contacts, combine the dynamics and the contact acceleration equations:

$$\begin{bmatrix} M(\mathbf{q}) & -J_c^{\mathsf{T}} \\ J_c & 0 \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{\nu}} \\ \mathbf{f}_c \end{bmatrix} = \begin{bmatrix} S^{\mathsf{T}}\boldsymbol{\tau} - \mathbf{h}(\mathbf{q}, \boldsymbol{\nu}) \\ -\dot{J}_c(\mathbf{q}, \boldsymbol{\nu})\, \boldsymbol{\nu} \end{bmatrix}. \tag{20}$$

The system solves for generalized accelerations and contact forces, subject to admissibility conditions (normal force, friction and non-penetration). Impacts and contact switching may require an impulse/contact solver. For an impact at velocity $\boldsymbol{\nu}^-$, the post-impact velocity $\boldsymbol{\nu}^+$ satisfies $M(\boldsymbol{\nu}^+ - \boldsymbol{\nu}^-) = J_c^{\mathsf{T}}\boldsymbol{\Lambda}$, $J_c\boldsymbol{\nu}^+ = 0$, where $\boldsymbol{\Lambda}$ is the contact impulse.

9. Computational workflow

- Build the kinematic tree and store each link's mass properties.
- Traverse from root to leaves to compute world poses (using $R(\boldsymbol{\xi}_b)$ for the base) and center-of-mass positions.
- Compute link and contact Jacobians with respect to $\boldsymbol{\nu}$.
- Assemble M and G; evaluate $C\boldsymbol{\nu}$ (often via recursive dynamics algorithms).
- Determine active contact modes and impose suitable constraints.
- Solve the constrained dynamics for $\dot{\boldsymbol{\nu}}$, validate contact forces, and integrate the state.

Integrating the state on the quaternion

The velocity is integrated as an ordinary vector; the configuration must respect $\|\boldsymbol{\xi}_b\| = 1$. A semi-implicit Euler step with time step Δt:

$$\boldsymbol{\nu}^+ = \boldsymbol{\nu} + \Delta t\, \dot{\boldsymbol{\nu}}, \tag{21}$$

$$\mathbf{p}_b^+ = \mathbf{p}_b + \Delta t\, \dot{\mathbf{p}}_b^+, \qquad \mathbf{q}_j^+ = \mathbf{q}_j + \Delta t\, \dot{\mathbf{q}}_j^+, \tag{22}$$

$$\boldsymbol{\xi}_b^+ = \mathrm{normalize}\big(\delta\boldsymbol{\xi} \otimes \boldsymbol{\xi}_b\big), \qquad \delta\boldsymbol{\xi} = \Big(\cos\tfrac{\|\boldsymbol{\phi}\|}{2},\, \sin\tfrac{\|\boldsymbol{\phi}\|}{2}\,\tfrac{\boldsymbol{\phi}}{\|\boldsymbol{\phi}\|}\Big), \quad \boldsymbol{\phi} = \Delta t\, \boldsymbol{\omega}_b^+. \tag{23}$$

Using the exact rotation $\delta\boldsymbol{\xi}$, rather than $\boldsymbol{\xi}_b + \Delta t\, \dot{\boldsymbol{\xi}}_b$, keeps the quaternion on the unit sphere up to round-off. The final normalization removes the remaining drift.

Practical implementations use the Composite Rigid Body Algorithm (CRBA) for M and the Recursive Newton–Euler Algorithm (RNEA) for $\mathbf{h} = C\boldsymbol{\nu} + G$. These are faster than the Jacobian sums above but give identical results, which makes the Jacobian-sum formulas a convenient reference for testing.

Correspondence with the code (`robot_model.py`)

Symbol	In code
$\mathbf{q} \in \mathbb{R}^{n+7}$, $\boldsymbol{\xi}_b$	`q`; `q[0:3]` $= \mathbf{p}_b$, `q[3:7]` $= \boldsymbol{\xi}_b = (w, x, y, z)$, `q[7:]` $= \mathbf{q}_j$
$\boldsymbol{\nu} \in \mathbb{R}^{n+6}$	`v`; `v[0:3]` $= \dot{\mathbf{p}}_b$, `v[3:6]` $= \boldsymbol{\omega}_b$ (both world frame), `v[6:]` $= \dot{\mathbf{q}}_j$
$R(\boldsymbol{\xi})$, Rodrigues, $\otimes$	`quat_to_R`, `axis_angle_to_R`, `quat_mul`
$(R_i, \mathbf{o}_i)$	`forward_kinematics(q)[link]` (link frame origin, *not* the CoM)
$\mathbf{p}_i$, $\sum m_i \mathbf{p}_i / \sum m_i$	`R @ link.com + o`; `center_of_mass(q)`
$[J_{v_i};\, J_{\omega_i}]$	`frame_jacobian(q, link, link.com)`: rows 0–2 $= J_{v_i}$, rows 3–5 $= J_{\omega_i}$
J_c (point contact)	`frame_jacobian(q, foot, corner)[0:3]`
$[\cdot]_\times$	`skew`
Integration (Sec. 9)	`integrate(q, v, dt)`
CoM velocity $J_{\mathrm{CoM}}\boldsymbol{\nu}$	`com_jacobian(q) @ v`
M, G, $C\boldsymbol{\nu}$, S	to be added in step 3

Key takeaway. With a quaternion base, separate the configuration $\mathbf{q} \in \mathbb{R}^{n+7}$ from the velocity $\boldsymbol{\nu} \in \mathbb{R}^{n+6}$. All Jacobians, the mass matrix, Coriolis and gravity terms and the contact equations live in the $(n+6)$-dimensional velocity space and keep exactly the same form as before, with simpler floating-base columns. The quaternion appears only in the rotation matrix $R(\boldsymbol{\xi}_b)$ and in the integration step. The mass matrix still comes from the link kinetic energies, gravity from the potential energy via virtual power, Coriolis and gyroscopic effects from the Newton–Euler equations of each link, and contact forces from virtual work.